EXCELLENCE

CLASS 11 - 2026-27

Physics | Unit 02

Units and Measurement

Principle of Homogeneity and Applications

By - Excellence Programme

7 October 2026

Topics to be covered

n_1 = old magnitude/value
u_1 = old unit

n_2 → new value

01 Warm-up theory: why n1 u1 = n2 u2

02 Unitless vs dimensionless (full theory)

03 Principle of homogeneity (paragraphs)

04 Finding dimensions of constants

05 Dimensionless functions (θ, sin, e^x, log)

06 Derive a formula + limitations

07 Board practice + CBSE write lines

Where we left Units 01

In Units 01 we learned that a physical quantity is something we can measure. Every measurement is written as a number × a unit. The size of the quantity itself does not change when we change the unit — only the number changes.

We also wrote dimensions using M, L and T. Today we use those dimensions as a tool: to check equations, to find unknown constants, and to derive the form of a formula.

Today: check equations · find unknown dimensions · derive formulas

Class notes will be saved after teaching. This deck is for the board.

What dimensional analysis is → [L] → m^3 ⇒ [MLT] ⇒ [$M^1L^3T^0$]

Dimensional analysis means studying a physical equation by looking only at the dimensions of each quantity. We do not put numbers first. We ask: do the powers of mass, length and time match on both sides?

Dim Speed = Dim Velocity = ?

If the dimensions do not match, the equation cannot be correct. If they do match, the equation is dimensionally correct. That does not always mean it is the true physical law — a wrong constant or a missing factor can still hide inside.

Dimensions are a filter. They catch many mistakes, but not every mistake.

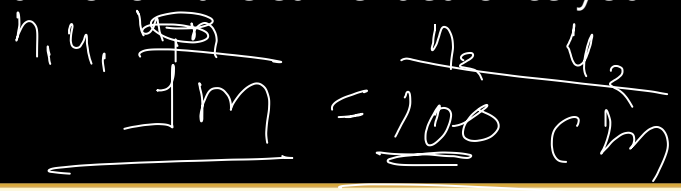
$m/s \Rightarrow [M^0L^1T^{-1}] = [M^0L^nT^{-1}]$

$\Rightarrow n = 1$

Why conversion works

The amount of a physical quantity is fixed. When we measure the same amount in a different unit, the number and the unit change together so that their product stays the same. That is why we write n1 u1 = n2 u2.

If the new unit is larger, the number becomes smaller. If the new unit is smaller, the number becomes larger. Density, force, torque — all follow this same idea once you know their dimensions.





n1 u1 = n2 u2

Rough work / extra work

Two conversion habits (continued)

1 m = 2 new units

Habit A — SI value given, new system asked. Keep the SI number. Replace each SI unit (kg, m, s, …) by how many new-system units fit into it. Then simplify. You are rewriting the same amount in new clothing.

Habit B — find how large one new unit is in SI. Build one new unit from the new mass, length and time using the dimensions of the quantity. Multiply those SI sizes. The result is the SI value of one new unit.

1 m = 2 k ⇒ 1 k. = $\frac{1}{2}$ m ⇒ 100 m ⇒ 100 m = 100 × 2 k = 200 k

out new unit ?

- Both habits use: the real amount does not change.
- Always write dimensions first for derived quantities (density, torque, …).

1 m = 2 new m

100 m

100 × (2 × new m) ⇒ 200 new m

Two conversion habits (board)

SI given → new asked

Write SI value. Replace each SI unit by the new unit.

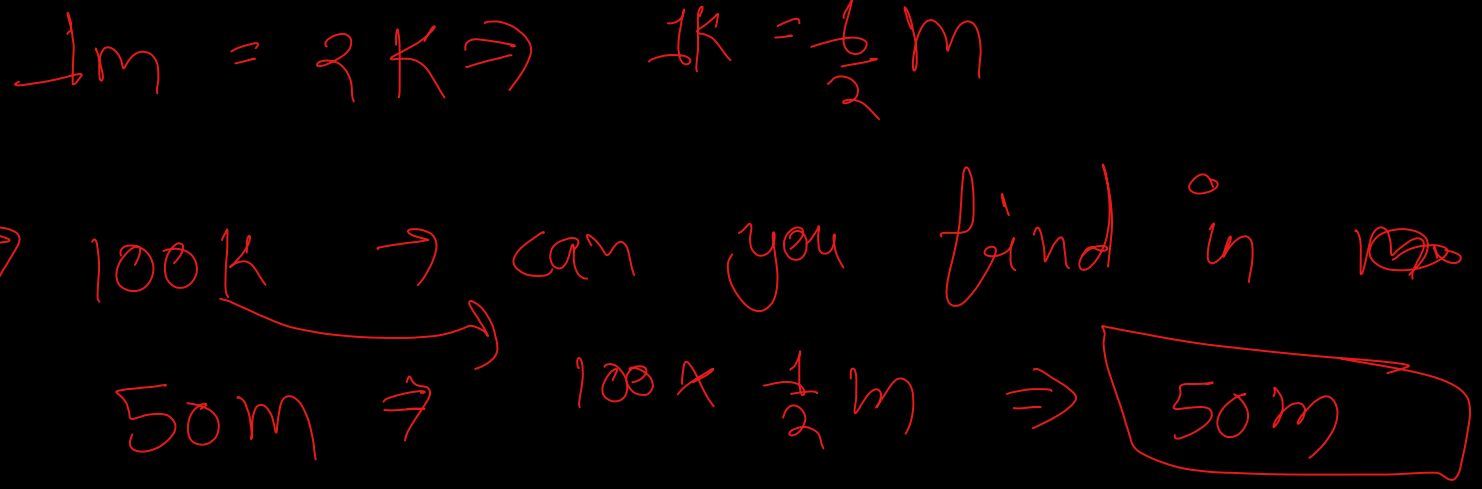


New unit value in SI

Write 1 new unit using M, L, T. Convert that to SI.

Both use the same idea: the actual amount does not change.

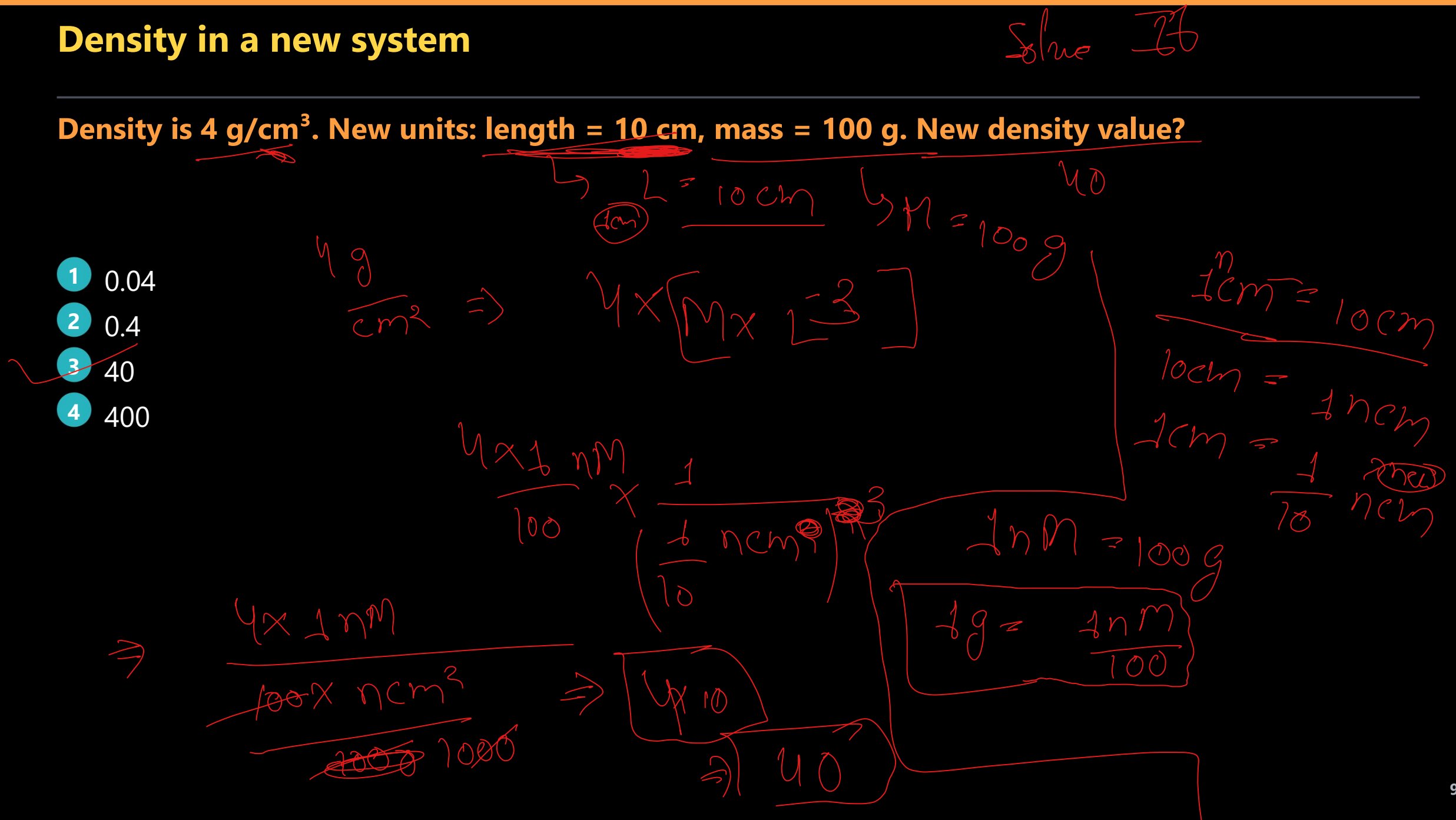
n1 u1 = n2 u2



Density in a new system

Density is 4 g/cm³. New units: length = 10 cm, mass = 100 g. New density value?

1. 0.04
2. 0.4
3. 40
4. 400



Solution: Density in a new system

Idea: rewrite 4 g/cm³ using the new mass and new length.

Step 1: New mass is bigger: 1 new mass = 100 g ⇒ 1 g = 1/100 new mass

Step 2: New length is bigger: 1 new length = 10 cm ⇒ 1 cm = 1/10 new length

Step 3: Density = mass / length³ = 4 × (1/100) ÷ (1/10)³

Step 4: = 4 × (1/100) × 1000 = 40

Final answer: 40

New unit of torque

Solve It

New system: length = 10 m, time = 2 s, mass = 5 kg. Find 1 new unit of torque in SI.

Torque = F × r ⇒ max[L] ⇒

↳ $[MLT^{-2}][L] \Rightarrow [ML^2T^{-2}]$

$L = 10m$; $T = 2S$; $M = 5kg$

$L' = \frac{L}{10}$ $T' = \frac{T}{2}$ $M'L'^2 = M'_S$

$ML^2T^{-2} \Rightarrow \frac{5M}{} \times \frac{10L^2}{} \times \left(\frac{2T}{}\right)^{-2} \Rightarrow \frac{ML^2T^{-2} \times 5 \times 100}{4}$

$\Rightarrow \frac{ML^2T^{-2} \times 5 \times 100}{4} \Rightarrow 125\,[ML^2T^{-2}]$

1. 125 N m
2. 0.125 N m
3. 8 N m
4. 8×10^{-3} N m

Solution: New unit of torque

Idea: one new torque unit = (new mass)(new length)^2 / (new time)^2.

Step 1: Torque has the same dimensions as energy: M L^2 T^{-2}

Step 2: Put the new values: 5 kg × (10 m)^2 / (2 s)^2

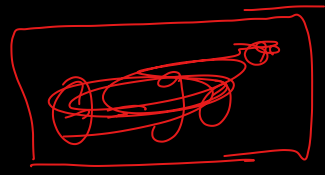
Step 3: = 5 × 100 / 4 = 125

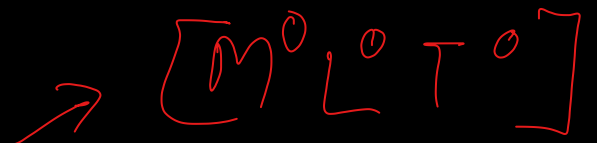
Final answer: 125 N m

Rough work / extra work

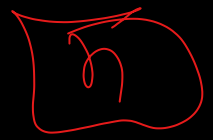
Unitless and dimensionless — two floors

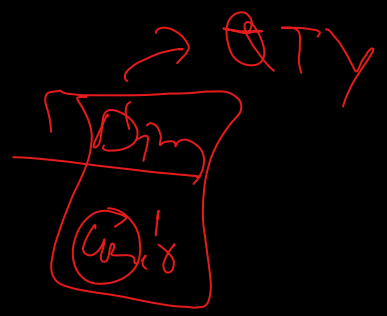
Students often mix two words: unitless and dimensionless. They sound alike, but they live on different floors of the same building. Floor 1 is the unit — the name we write after the number. Floor 2 is the dimension — the powers of M, L and T hidden inside the quantity.



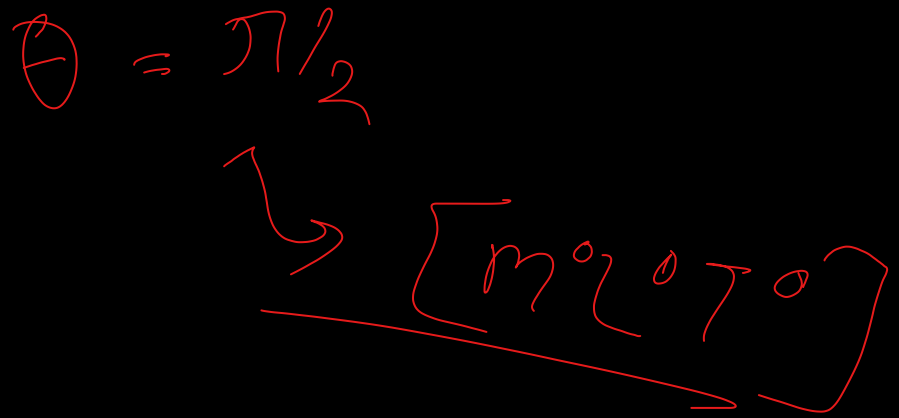


Unitless means we do not write a unit after the number. Dimensionless means every power is zero: $M^0\ L^0\ T^0$. These two ideas can travel together, but they are not forced to stay together.





Learn both words separately. Do not treat them as synonyms.



Unitless and dimensionless (continued)

If a quantity is unitless, it must also be dimensionless. There is nowhere for a hidden M, L or T to hide if there is no unit at all. Strain and refractive index are common examples: they are pure numbers.

The reverse is not always true. A quantity can be dimensionless and still have a unit. Plane angle is the best school example. Its dimension is none, but its SI unit is radian. Degree is another unit of the same dimensionless quantity.

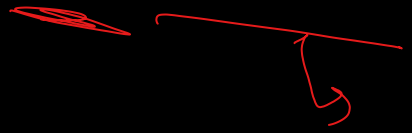
Angle has a unit (radian) but no dimension.

What follows from what

Has dimension ⇒ must have a unit. If mass, length or time appear in the formula, we need a unit to measure that quantity. You cannot measure force without newton (or dyne, or some other force unit).

Has a unit ⇒ may or may not have a dimension. Angle has a unit. Force has a unit. Only force has a dimension in the M-L-T sense. So writing a unit does not automatically create a dimension.

- Unitless ⇒ dimensionless (always).
- Dimension ⇒ unit (always).
- Unit ⇒ dimension? Not always. Angle is the counter-example.



True / false lines (copy)

1. A unitless quantity may have dimension. → False
2. A dimensionless quantity may be unitless. → True (μ, strain)
3. A quantity that has a unit must have a dimension. → False (angle)
4. A quantity that has a dimension must have a unit. → True



Has dimension ⇒ must have unit. Unitless ⇒ must be dimensionless.

Best CBSE example: plane angle. Unit = radian. Dimension = none.

Rough work / extra work

Decision table (easy)

If you know...	Then...
no unit (unitless)	must be dimensionless
has a unit	may or may not have dimension
no dimension	may or may not have a unit
has dimension	must have a unit

Floor 1 = unit. Floor 2 = dimension. Unit can exist without dimension.



Principle of homogeneity — meaning

The principle of homogeneity says that in any correct physical equation, every term that is added, subtracted or compared must have the same dimensions. Left side and right side must also match. Nature does not add metres to kilograms.

Think of dimensions as the "type" of a quantity. You may multiply or divide different types and get a new type. You may not add different types. That is the whole rule in one sentence.

In any correct equation, every term must have the same dimensions.

Principle of homogeneity (continued)

Why does this rule exist? Because an equation is a statement about equality of physical amounts. Two sides can be equal only if they are the same kind of thing. Force equals mass × acceleration — both sides are M L T^{-2}. Force cannot equal pressure, because pressure is force per area.

$F \neq \frac{F}{A} \Rightarrow A = 1 \Rightarrow M^2 = 1$ (Invalid dimension)

Multiplication and division are allowed for unlike dimensions because they create a new quantity. Speed is length divided by time. Density is mass divided by volume. The new quantity simply carries the combined dimensions.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

4 kg + 8 m is meaningless. 5 m / 2 s = 2.5 m/s is fine.

$\rightarrow 2.5\ \frac{m}{s}$

$\rightarrow$ ~~12 (kg + m)~~ not valid

What homogeneity allows and forbids

Allowed: add or subtract quantities with the same dimensions. Compare them with > , < or = . Multiply or divide any dimensions. Raise a quantity to a power (the dimensions take that power).

Forbidden: add force to pressure, add energy to torque as if they were different physical ideas that somehow mix, or write an equation where one side is time and the other is length. Same dimensions do not prove same quantity — work and torque share $M\,L^2\,T^{-2}$ but they are not the same.

- Same dimensions ⇒ may add / subtract / compare.
- Same dimensions ≠ same physical quantity.

Principle of homogeneity (3 marks)

Definition: The dimensions of each term on both sides of a physical equation must be the same. This is called the principle of homogeneity of dimensions.

$F = ma$

$[MLT^{-2}] = [M][LT^{-2}]$

$= [MLT^{-2}]$

Use 1: Check if a formula is dimensionally correct.

Use 2: Find dimensions of unknown constants in an equation.

Use 3: Derive the form of a formula (pendulum, later chapters).

Same dimensions on every + or − term. That is the whole idea.

Dimensionally correct does not always mean physically correct.

When you see $+, -, <, >, =$ Dim of Terms on both side ~~will be~~ must be same

Rough work / extra work

Board shortcut for checking an equation

When you only want to check dimensions, you may mentally replace +, −, > and < by = . You are not changing the real maths. You are only saying: every piece that sits in those places must carry the same dimensions.

Example: $v = u + a\,t$. For dimensions, write $[v] = [u] = [a\,t]$. Check each piece. Velocity, initial velocity, and acceleration × time must all be $L\,T^{-1}$.

For dimension check only: treat +, −, > , < like =

$[M L T] + [M L T^2][T][T^{-1}] \Rightarrow$ valid

Just replace + with = $\Rightarrow [MKT] = [MLT^2] \Rightarrow$ $F = TR$ theorem

What you can and cannot do

Allowed

Add / subtract same dimensions. Multiply / divide any.

Not allowed

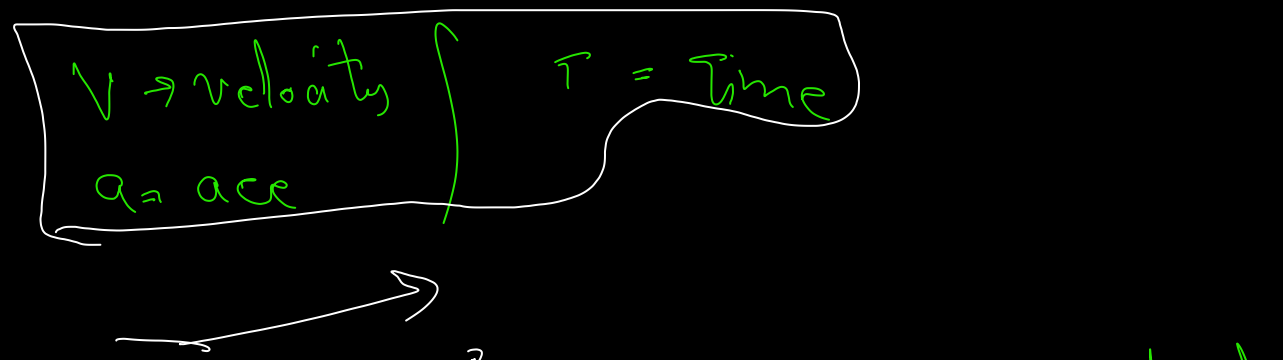
Add force to pressure. Compare 4 kg with 2 m.

Displace = length / Time = L / T

V = L T⁻¹

4 kg + 8 kg = 12 kg. 3 kg / 5 m = 0.6 kg/m (linear density).

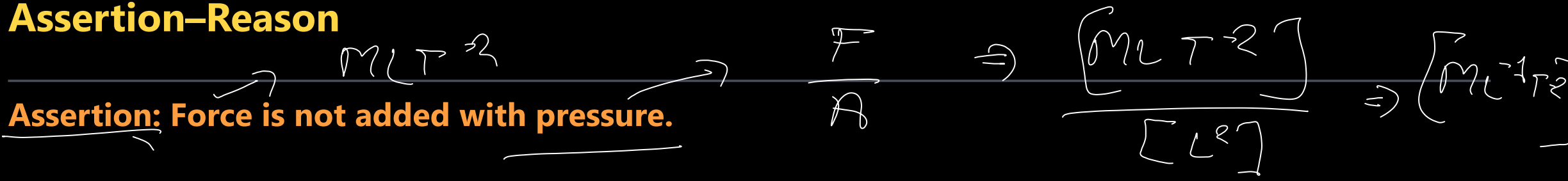
Different names can share dimensions. Same dimensions do not mean same quantity.



$F \pm \frac{V}{T} \times m = 0$ → valid

Time [T]

Assertion–Reason



Assertion: Force is not added with pressure.

Reason: Quantities with different dimensions are not added or subtracted.

1. Both true, Reason explains Assertion
2. Both true, Reason does not explain
3. Assertion true, Reason false
4. Both false

Solution: Assertion–Reason

Check Assertion first, then Reason, then the link.

Step 1: Force dimension = M L T^{-2}. Pressure = F/A = M L^{-1} T^{-2}.

Step 2: These are different, so force and pressure cannot be added. Assertion true.

Step 3: Reason says the same rule. So Reason is true and it explains Assertion.

Final answer: Option 1 — both true, Reason explains Assertion

Finding dimensions of unknown constants

Often an equation contains letters a, b, α, β whose dimensions are not given. Homogeneity still works. Wherever two things are added or subtracted, their dimensions match. Wherever something equals the left side, it matches the left side.

So the equation itself becomes the teacher. You do not invent a new formula. You read the structure: who is added to whom, who is multiplied by whom, and who sits alone on one side.

If A + B = C, then [A] = [B] = [C]

$20mt + 30m = 50m$

Rough work / extra work

Finding constants (method continued)

Step by step method. First mark every + and −. Those pieces must have equal dimensions. Next equate each such piece with the left-hand side if the whole sum equals the left side. Finally solve for the unknown using known dimensions of force, pressure, velocity, and so on.

If a quantity is divided or multiplied, rearrange. Example: if $F = x / (b\ t^2)$, then $b = x / (F\ t^2)$. Write dimensions of the right side and simplify. Pure numbers like 2 or π do not change dimensions.

Do not invent formulas. Use only what the equation itself tells you.

Find dimensions — board steps

Step 1: Mark every + or −. Those pieces must match.

Step 2: Equate each piece with the left side (or with its neighbour).

Step 3: Solve for the unknown using known dimensions.

If A + B = C, then [A] = [B] = [C]

Do not invent formulas. Use only what the equation itself tells you.

Electric field constants

$\vec{E} = (A / x^2)\,\hat{i} + (B / y^3)\,\hat{j}$. SI units of A and B are

$\frac{N}{C} = \frac{A}{x^2}\hat{i} + \frac{B}{y^3}\hat{j}$ ⇒ By Principal of homogeneity

1. $N\,m^3\,C^{-1}$, $N\,m^2\,C^{-1}$
2. $N\,m^2\,C^{-1}$, $N\,m^3\,C^{-1}$
3. $N\,m^3\,C$, $N\,m^2\,C$
4. $N\,m^2\,C$, $N\,m^3\,C$

$\frac{N}{C} = \left[\frac{A}{x^2}\right] = \left[\frac{B}{y^3}\right]$

$\frac{N}{C} = \frac{A}{L^2}$

$A = NC^{-1}L^2$

$\frac{N}{C} = \frac{B}{y^3}$

$B = NC^{-1}L^3$

Solution: Electric field constants

Idea: each term on the right must have the same unit as E.

Step 1: Electric field E = force / charge, so unit of E = N/C

Step 2: $A / x^2 = E \Rightarrow A = E \times x^2 = (N/C) \times m^2$

Step 3: $B / y^3 = E \Rightarrow B = E \times y^3 = (N/C) \times m^3$

Final answer: $A \rightarrow N\,m^2\,C^{-1}$, $B \rightarrow N\,m^3\,C^{-1}$

Pressure-style equation

$P = \frac{F}{A} \rightarrow [M^1 L^{-1} T^{-2}] [L^2] \Rightarrow [ML^{-1}T^{-2}]$

$(P + b / V^2) = a$. P is pressure, V is volume. Find [a] and [b].

$P = [ML^{-1}T^{-2}]$

$a = M^0 L^{-1} T^{-2}$

$b = M^0 L^5 T^{-2}$

$[P] = \left[\frac{b}{V^2}\right] = [a]$

$[ML^{-1}T^{-2}] = [a] \Rightarrow a = [ML^{-1}T^{-2}]$

$\frac{b}{V^2} = ML^{-1}T^{-2} =$

$\frac{b}{L^6} \Rightarrow ML^{-1}T^{-2} \Rightarrow b = ML^5 T^{-2}$

Solution: Pressure-style equation

Idea: things added must match. Right side must match left side.

Step 1: a sits alone on the right, so $[a] = [P] = M\,L^{-1}\,T^{-2}$

Step 2: P and b/V^2 are added, so $[b/V^2] = [P]$

Step 3: $[b] = [P] \times [V]^2 = M\,L^{-1}\,T^{-2} \times L^6 = M\,L^5\,T^{-2}$

Final answer: $[a] = M\,L^{-1}\,T^{-2}$, $[b] = M\,L^5\,T^{-2}$

Rough work / extra work

$v = \alpha t + \beta x - \gamma x t$

$[v] \rightarrow [L]$

$[t] \Rightarrow [T]$

If velocity $v = \alpha t + \beta x - \gamma x t$, find dimensions of α, β and γ.

$\alpha \rightarrow LT^{-2}$

$\beta \rightarrow T^{-1}$

$\gamma \rightarrow T^{-2}$

Solution: $v = \alpha t + \beta x - \gamma x t$

Idea: treat each term like it equals velocity. $[v] = L\,T^{-1}$.

Step 1: $\alpha \times \text{time} = \text{velocity} \Rightarrow [\alpha] = \text{velocity} / \text{time} = L\,T^{-2}$

Step 2: $\beta \times \text{length} = \text{velocity} \Rightarrow [\beta] = \text{velocity} / \text{length} = T^{-1}$

Step 3: $\gamma \times \text{length} \times \text{time} = \text{velocity} \Rightarrow [\gamma] = T^{-2}$

Final answer: $[\alpha] = L\,T^{-2}$, $[\beta] = T^{-1}$, $[\gamma] = T^{-2}$

Force formula $F = (a - x) / (b\ t^2)$

Equation of force: $F = (a - x)/(b\ t^2)$. Find dimensions of a and b.

Solution: Force formula $F = (a - x) / (b\ t^2)$

Idea: first use the subtraction, then rearrange for b.

Step 1: $a - x$ means $[a] = [x]$. Distance x is length, so $[a] = L$

Step 2: From $F = x / (b\ t^2)$, we get $b = x / (F\ t^2)$

Step 3: $[b] = L / (M\ L\ T^{-2} \times T^2) = 1/M = M^{-1}$

Final answer: $[a] = L$, $[b] = M^{-1}$

Rough work / extra work

Dimensionless functions — why they matter

Trigonometric functions, exponential and logarithm appear often in waves, SHM and decay. For dimensions we treat them in a special way. Their values are pure numbers, so they do not add M, L or T of their own.

→ There is a strict rule: the thing sitting inside sin, cos, tan, e , or log( ) must itself be dimensionless. You cannot take sin of a length or log of a force. The argument must reduce to $M^0\ L^0\ T^0$.

$\sin(x) \Rightarrow$ what is dim of x = ? If is dimensionless

Inside sin, cos, tan, exp, log — that argument must be dimensionless.

Dimensionless functions (continued)

Angle θ is dimensionless because θ = arc length / radius = L/L. So [θ] = 1. Then sin θ, cos θ and tan θ are ratios of sides of a triangle — again length over length — so they are dimensionless too.

$$\sin\theta = \frac{P}{H} = \frac{\lambda}{\lambda} = 1 \text{ unitless } [L^0]$$

When you only check dimensions, write θ, sin θ, cos θ, e^x and log x as the number 1. They do not change the dimensions of whatever they multiply. If P = α sin β, then [α] = [P] and [β] = 1.

For dimensions: write θ, sin θ, e^x, log x as 1

Why angle is dimensionless (2 marks)

H.W

Plane angle is defined as the ratio of arc length to radius. Both arc and radius are lengths, so their ratio has no dimension: $\theta = L/L = 1$. Therefore $[\theta] = M^0 L^0 T^0$.

Still, angle has a unit: radian (or°).

Dimensionless does not mean no unit

Same idea: strain = $\Delta L / L$. Dimensionless. Usually unitless too.

$P = \alpha \sin \beta$

Dim of α & β ?

If momentum $P = \alpha \sin \beta$, find dimensions of α and β.

$$[P] = [MLT^{-1}] = [\alpha]$$

$$\alpha = MLT^{-1}$$

Solution: $P = \alpha \sin \beta$

Idea: sin(anything) is just a pure number for dimensions.

Step 1: Inside sin must be dimensionless. So $[\beta] = M^0 L^0 T^0 = 1$

Step 2: $\sin \beta = 1$ (dimensionally). So $P = \alpha \times 1 \Rightarrow [\alpha] = [P]$

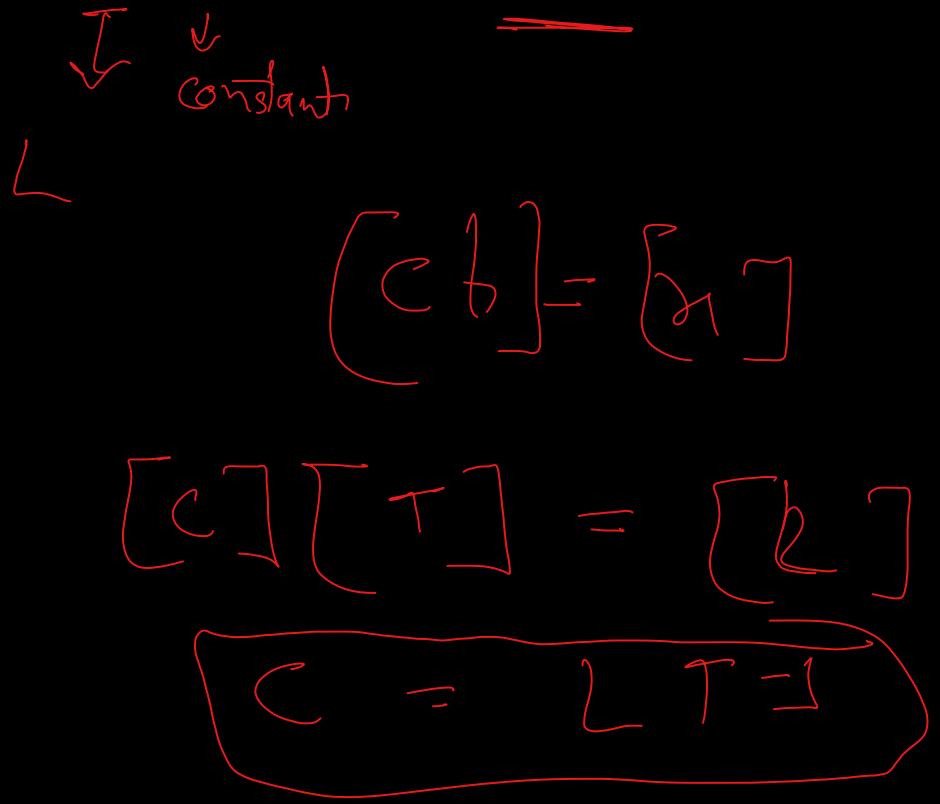
Step 3: Momentum $P = m v$, so $[\alpha] = M L T^{-1}$

Final answer: $[\alpha] = M L T^{-1}$, $[\beta]$ = dimensionless

Rough work / extra work

Wave equation: find [c]

Y = A sin[(2π/λ)(c t – x)]. Find dimension of c.



Solution: Wave equation: find [c]

Idea: look only at (c t − x). Those two must match.

Step 1: c t and x are subtracted, so [c t] = [x]

Step 2: [c] × time = length ⇒ [c] = length / time = L T^{-1}

Final answer: [c] = L T^{-1} (like speed)

Using dimensions to derive a formula

Sometimes we know which quantities a formula depends on, but we do not know the powers. Dimensional analysis can find those powers. We assume a product form, write dimensions on both sides, and match the powers of M, L and T one by one.

Example idea: time period T of a simple pendulum depends on length l and acceleration due to gravity g. We write $T = k\, l^a g^b$, where k is a pure number. Then we find a and b. The method cannot give the value of k — experiment or deeper theory gives $k = 2\pi$.

$T \propto l^a g^b$ → find a and b by comparing powers

Derive formula — what we assume

We assume the unknown relation is a product of powers: quantity = constant × (first quantity)^a × (second)^b × …. This works when each variable appears once as a power. It fails if the true law has a sum of two different terms, or a trigonometric factor that we forgot to treat as dimensionless.

Also, if a quantity depends on more variables than we can balance with M, L, T alone, dimensions may not give a unique answer. Then we need more physics. Keep that honesty for board answers under "limitations".

k is a pure number here. Dimensions ignore it.

Pendulum step by step

Write $T = k\, l^a\, g^b$. Now match dimensions of both sides.

Left side $[T] = T^1$. Right side $= L^a \times (L\,T^{-2})^b$

Right side becomes $L^{a+b}\,T^{-2b}$

Match time power: $1 = -2b \Rightarrow b = -1/2$

Match length power: $0 = a + b \Rightarrow a = +1/2$

So $T = k\,\sqrt{(l/g)}$. Real formula uses $k = 2\pi$.

Rough work / extra work

Uses of dimensional analysis

Dimensional analysis has three standard school uses. First, check whether an equation is dimensionally correct. Second, convert a numerical value from one system of units to another using n1 u1 = n2 u2. Third, find how physical quantities are related — the powers in a formula — when we know which quantities matter.

For a 3-mark CBSE answer, write the definition of homogeneity briefly, list these three uses, and add one limitation. That structure scores cleanly without long English.

Check · Convert · Derive

Limitations of dimensional analysis

Dimensional analysis cannot find the value of a dimensionless constant such as 2π, 1/2 or e. Those numbers have no M, L or T, so dimensions never see them. The real formula must come from experiment or from a fuller derivation.

It also cannot decide between two formulas that already have the same dimensions. And if a quantity depends on more independent variables than the number of base dimensions we use, we may not get unique powers. Write these limits honestly in board answers.

- Cannot give pure numbers like 2π.
- Cannot prove a formula is physically true — only dimensionally possible.
- Fails when too many variables or when the law is not a simple product.

Uses of dimensional analysis (3 marks)

1. To check whether an equation is dimensionally correct.

2. To convert a value from one system of units to another.

3. To find the relation between physical quantities (derive formula).

Check · Convert · Derive

Limitation: cannot find dimensionless constants like 2π or 1/2.

NEET: time period from α and β

Max acceleration of oscillating particle is α, max velocity is β. Time period is

1. $2\pi\ \alpha/\beta$
2. $2\pi\ \alpha\ \beta$
3. $2\pi\ \beta/\alpha$
4. $2\pi\ \alpha^2\ /\ \beta^2$

Solution: NEET: time period from α and β

Easy way: check which option has dimension of time.

Step 1: In SHM, max speed $\beta = A\omega$, max acceleration $\alpha = A\omega^2$

Step 2: Divide: $\alpha/\beta = \omega$. Time period is $T = 2\pi / \omega$

Step 3: So $T = 2\pi \times (\beta/\alpha)$

Final answer: $2\pi\ \beta/\alpha$ (option 3)

Rough work / extra work

Which does NOT have dimension of time?

L, C, R are inductance, capacitance, resistance. Which has no dimension of time?

1. RC
2. L/R
3. √(LC)
4. L/C

Solution: Which does NOT have dimension of time?

Remember three time formulas from circuits.

Step 1: RC has dimension of time. L/R has dimension of time.

Step 2: √(LC) also has dimension of time.

Step 3: L/C does not become time. That is the odd one out.

Final answer: L/C (option 4)

Units 02 — theory in one place

Homogeneity: every term that is added, subtracted or compared must have the same dimensions. Unitless always means dimensionless; having a dimension always means having a unit; having a unit does not always mean having a dimension (angle).

For dimensions, treat θ, sin θ, e^x and log x as 1. Uses of dimensional analysis: check a formula, convert units, derive a relation. Limitation: cannot find pure numbers like 2π, and cannot prove a formula is physically true.

Lock these four ideas before board practice.

Units 02 lock lines

Homogeneity: every + / − / compare term has the same dimensions.

Unitless ⇒ dimensionless. Has dimension ⇒ must have unit.

θ, sin θ, e^x, log x → treat as 1 for dimensions.

Uses: check formula, convert units, derive relation.

Limitation: cannot get pure numbers like 2π.

Ready theory lines

1. Principle of homogeneity: dimensions of all terms in an equation are equal.

2. Only quantities of the same dimensions can be added or subtracted.

3. Angle is dimensionless but has unit radian.

4. Arguments of sin, cos, tan, log and exp must be dimensionless.

5. Dimensional analysis cannot give the value of a dimensionless constant.

For 3 marks: definition + one use + one limitation.

Rough work / extra work

Practice

1. If $F = a / t + b x^2$, find [a] and [b].

2. If $y = A \cos(\omega t - kx)$, show $[\omega]/[k] = L T^{-1}$.

3. Write three uses and two limitations of dimensional analysis.

4. Derive $T \propto \sqrt{(l/g)}$ for a simple pendulum.

5. True/False: A quantity with a unit must have a dimension.

6. If $E = \alpha / \beta$, and E is energy, β is time, find [α].

Next: significant figures and errors (PW later lectures).